GASLA-18 Conference

Strength of *wh*-island effects in subject vs. object relatives for L1/L2 English adults

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fredzenker.github.io/gasla2026.pdf

Two types of relative clause (RC)

1. **Gapped** RCs

(1) the officers$_1$ [that __$_1$ arrested the criminal]

- The primary relativization type in English

2. **Resumptive** RCs

(2) *the officers$_1$ [that **they**$_1$ arrested the criminal]

- Widely held to be ungrammatical in all English RCs
- Used where gapped RCs **difficult to process** or **inadmissible**

(e.g., Ferreira & Swets, 2005; Morgan & Wagers, 2018; Ross, 1967; Shlonsky, 1992)

RC processing asymmetries in L1-English adults

- Short-distance: SRC > ORC

(3) a. the officer$_1$ [that ___$_1$ arrested the criminal] (SRC) >
b. the criminal$_1$ [that the officer arrested ___$_1$] (ORC)

(e.g., Just et al., 1996; King & Just, 1991)

- Long-distance: ORC > SRC

(4) a. *the intern$_1$ [that I wonder [who$_2$ ___$_2$ hired ___$_1$]] (ORC) >
b. *the company$_1$ [that I wonder [who$_2$ ___$_1$ hired ___$_2$]] (SRC)

(e.g., Han et al., 2012; Keffala & Goodall, 2011; Morgan & Wagers, 2018)

Morgan and Wagers (2018)

Written elicited production task

(5) *Whether*-island conditions with **gapped** vs. **resumptive** RC

BASE: The ogre doesn't care if a prince slayed the troll.

PROMPT: There's a price$_1$ that...

a. the ogre doesn't care [if the troll slayed {*__$_1$ /***him**$_1$}]. (ORC) >

b. the ogre doesn't care [if {*__$_1$ /***he**$_1$} slayed the troll]. (SRC)

- More resumption in SRC islands than in ORC islands

Acceptability judgment task

- Gapped RCs rated lower in SRC islands than in ORC islands

Conclusion: SRC islands stronger than ORC islands

The present investigation

- Part of a larger dissertation project
- Inspired in part by Morgan and Wagers's (2018) design
- Focus: subject-object asymmetry in argument *wh*-island RCs
  - Not previously looked at with adult L2 learners of English

Participant groups

Adult L1-English controls (ENSs)

Adult L1-Korean L2 learners of English (KLEs)

Adult L1-Mandarin L2 learners of English (MLEs)

Why KLEs and MLEs?

- Korean and Mandarin lack overt *wh*-movement in *wh*-questions
- Korean and Mandarin differ in the distribution of grammatical resumption in RCs

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan 2016)

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan 2016)

English & Korean: Only gapped RCs in the short-distance and long-distance environments tested

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan 2016)

Mandarin: Optional resumption in 3/4 of the same environments

English, Korean, and Mandarin compared

Grammatical gapped (–) and resumptive (+) RCs in environments tested

	Short-distance		Long-distance		*Wh*-island	
Language	SRC	ORC	SRC	ORC	SRC	ORC
English	–	–	–	–	Neither	Neither
Korean	–	–	–	–	N/A	N/A
Mandarin	–	±	±	±	N/A	N/A

Note. Based on available literature and judgments of our language consultants.

(Han, 2013; Keenan & Comrie, 1979; Morgan & Wagers, 2018; Pan 2016)

Wh-islands: English allows neither gapped RCs not resumptive RCs; that environment does not exist in Korean and Mandarin

Research questions (RQs)

RQ1: Does the subject–object asymmetry in the strength of argument *wh*-islands hold for adult L2 learners of English whose L1 lacks *wh*-movement in *wh*-questions?

RQ2: Is the strength of the asymmetry modulated by L2 English proficiency?

Participant background information

Participants distributed across two sub-studies: an ORC study and an SRC study

Participants in ORC and SRC studies prior to exclusions

Study	Group	*n*	Age at testing	C-test score	Age of onset
ORC	ENS	90	29.98 (18–71)	42.74 (26–49)	—
ORC	KLE	69	26.29 (18–41)	29.29 (7–45)	9.35 (8–15)
ORC	MLE	76	28.14 (18–45)	28.85 (7–49)	9.71 (8–14)
SRC	ENS	61	26.43 (18–58)	42.85 (22–50)	—
SRC	KLE	66	25.38 (19–36)	32.05 (11–46)	9.23 (8–13)
SRC	MLE	73	29.51 (18–46)	28.80 (11–48)	10.29 (8–14)

Note. Values are means and ranges; the C-test (max = 50) tested English ability.

Sequencing of tasks

1. Consent form
2. Language background questionnaire
3. **Oral elicited production task**
4. Self-paced reading task
5. **English acceptability judgment task**
6. Korean/Mandarin acceptability judgment task (L2 groups only)
7. English proficiency C-test (Zenker, 2024)
8. Exit survey

Oral elicited production tasks (EPTs)

- Conditions eliciting short-distance, long-distance, and *wh*-island RCs
- 15 critical items (3 conditions × 5 tokens, Latin-squared)
- 15 fillers eliciting short-distance SRCs with biclausal sentences

_Wh_-island conditions in the EPTs

Study	Target response with **gapped** vs. **resumptive** RC
ORC	the man$_1$ [that Mary wonders [which officers$_2$ __$_2$ arrested {*__$_1$/***him**$_1$} last week]]
SRC	the man$_1$ [that Mary wonders [which book$_2$ {*__$_1$/*he$_1$} borrowed __$_2$ last month]]

EPT example trial in the SRC *wh*-island condition

Mary wonders which book this man borrowed last month. ← stimulus sentence

Lisa wonders which book this man borrowed last month. ← foil sentence

Which man is this? ← question

Lisa wonders which book this man borrowed last month.

Recording...

See Ferreira and Swets (2005) for a similar procedure

"the man that Mary wonders which book {__/**he**} borrowed last month"

EPT response categories

1. Gapped RC
2. Resumptive RC
3. Other RC
4. Nontarget

Some other RC responses:

- "the man that is wondered by Mary which officers arrested last week"
- "the man who borrowed a book that Mary wonders about"

Some nontarget responses:

- "Mary wonders which officers arrest this man last week"
- "Mary wonders which book he borrowed last month"

EPT data exclusions

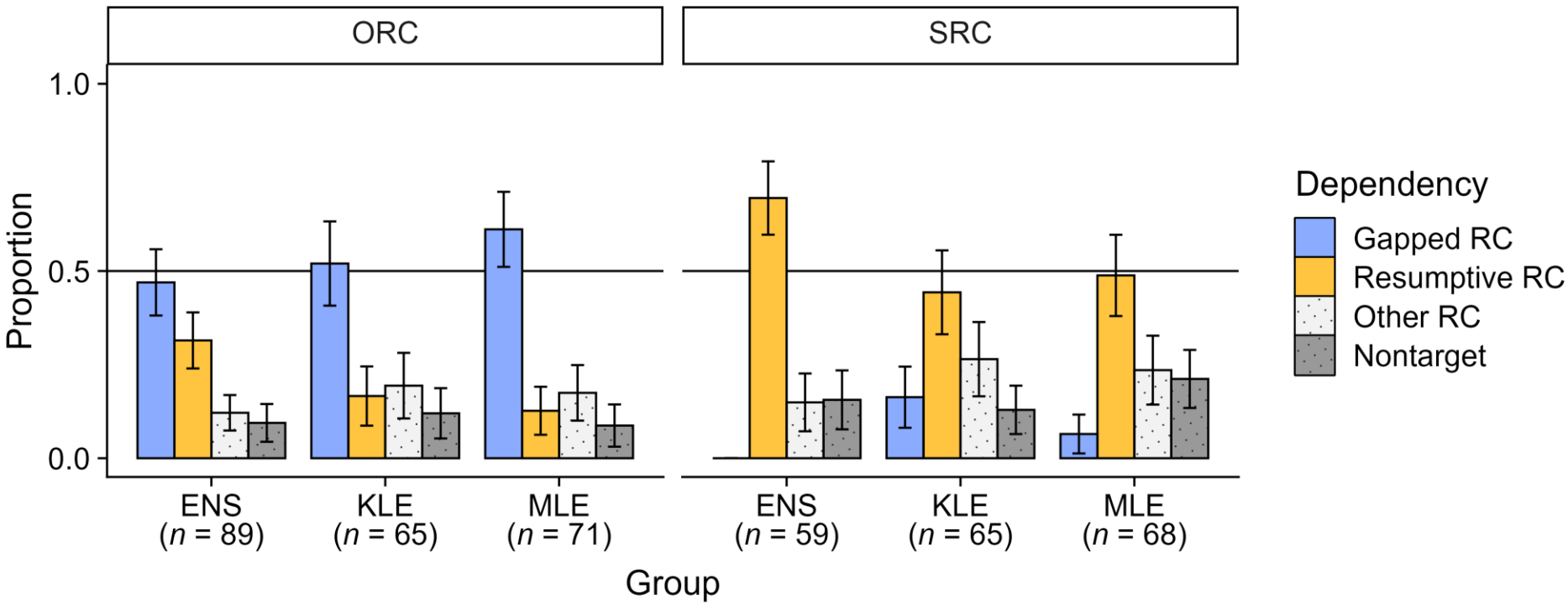
Experimental error

- ORC study: N/A
- SRC study: 1 MLE

Participants with only nontarget responses

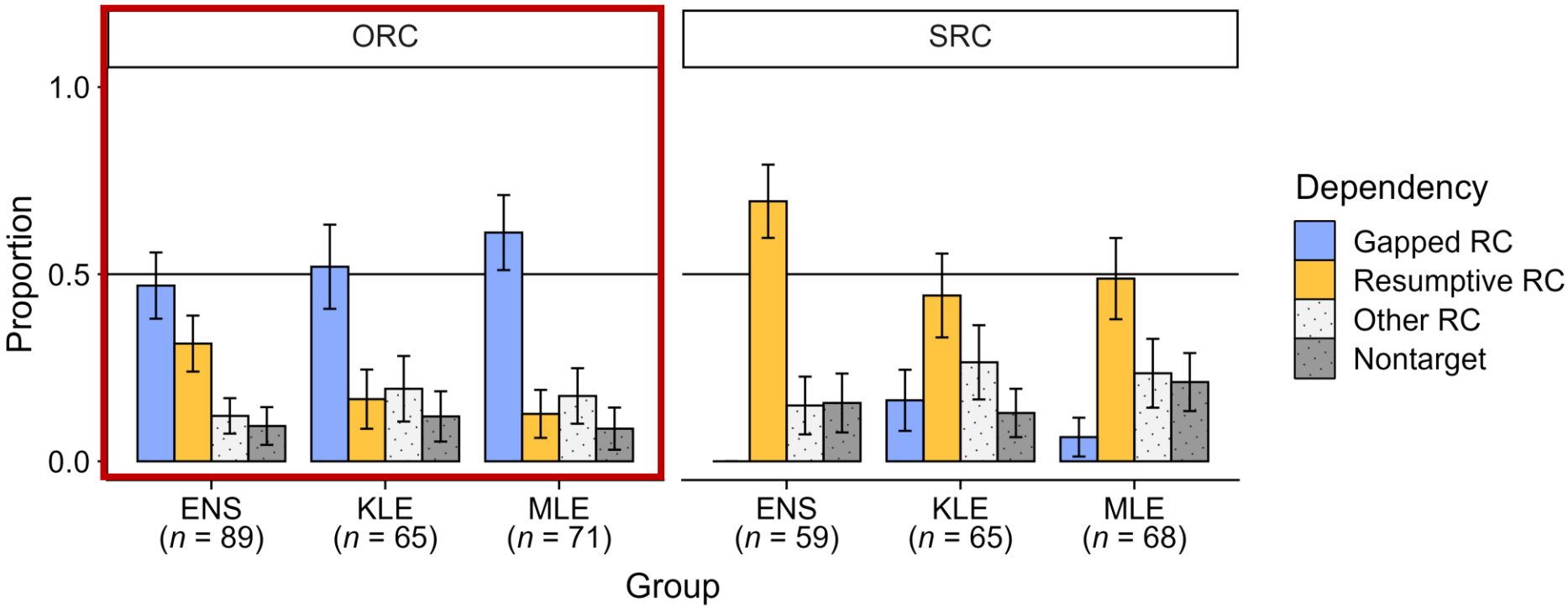
- ORC study: 1 ENS, 4 KLEs, 5 MLEs
- SRC study: 2 ENSs, 1 KLE, 4 MLEs

EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group + (1 | Participant) + (1 | Item).

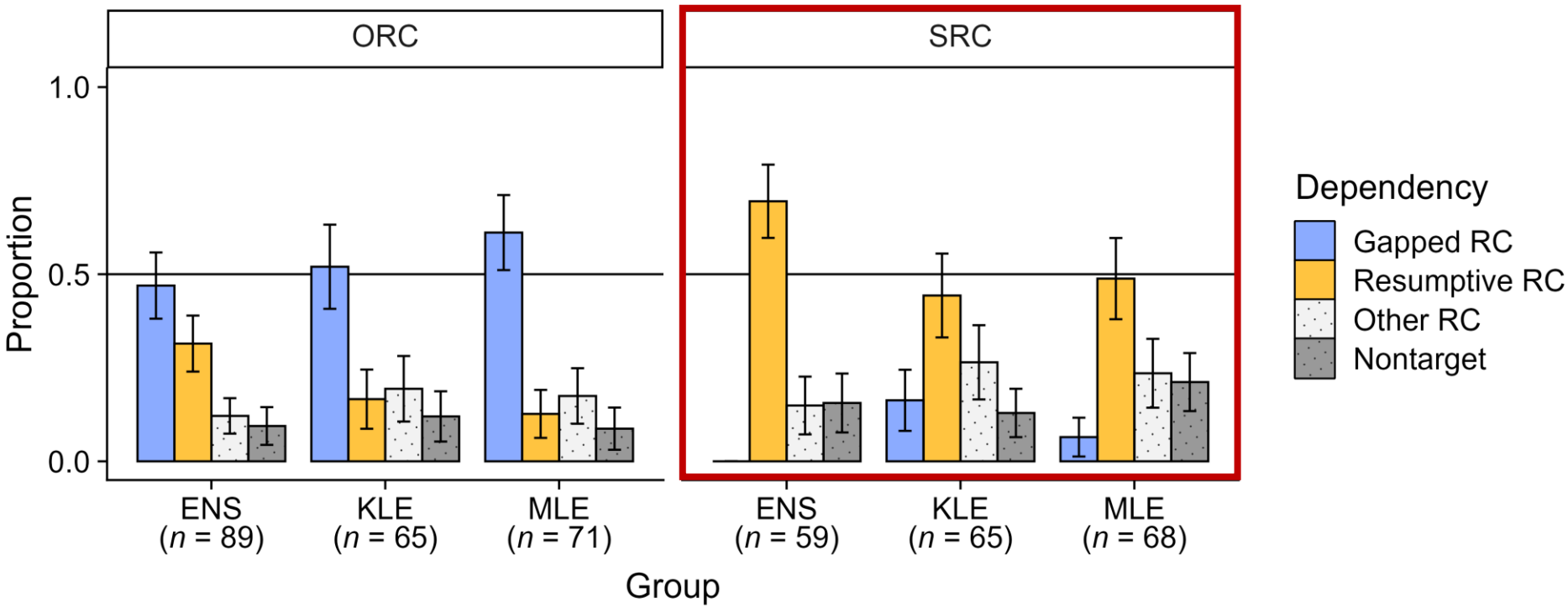
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group + (1 | Participant) + (1 | Item).

ORC study: Gapped RCs most common

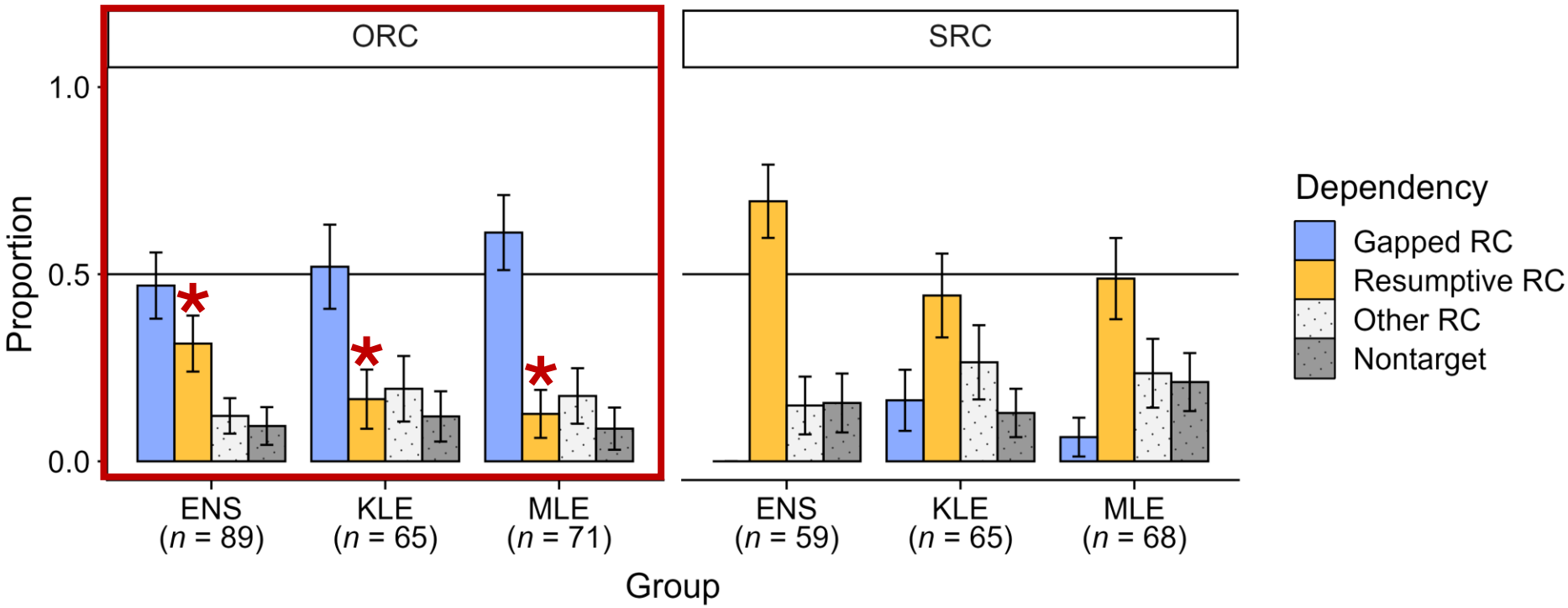
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group + (1 | Participant) + (1 | Item).

SRC study: Resumptive RCs most common; gapped RCs least common

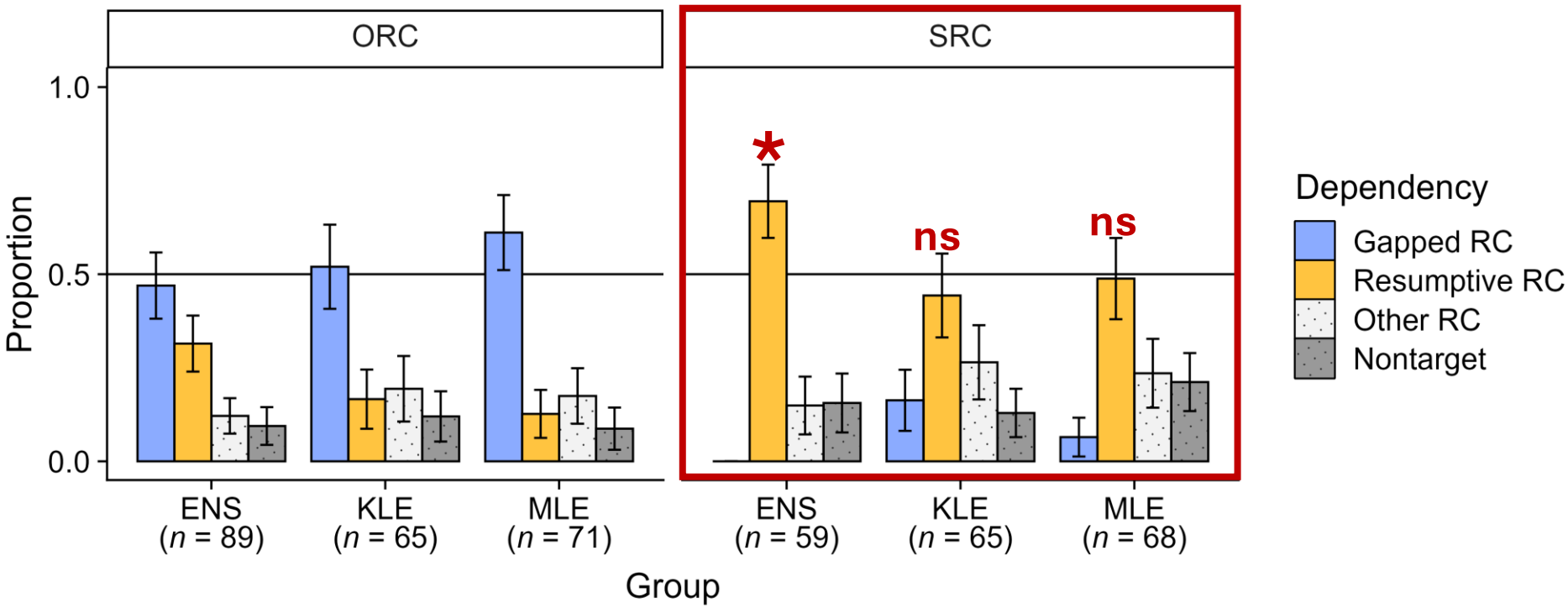
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group + (1 | Participant) + (1 | Item).

ORC study: Resumptive RCs significantly less common than other response types combined

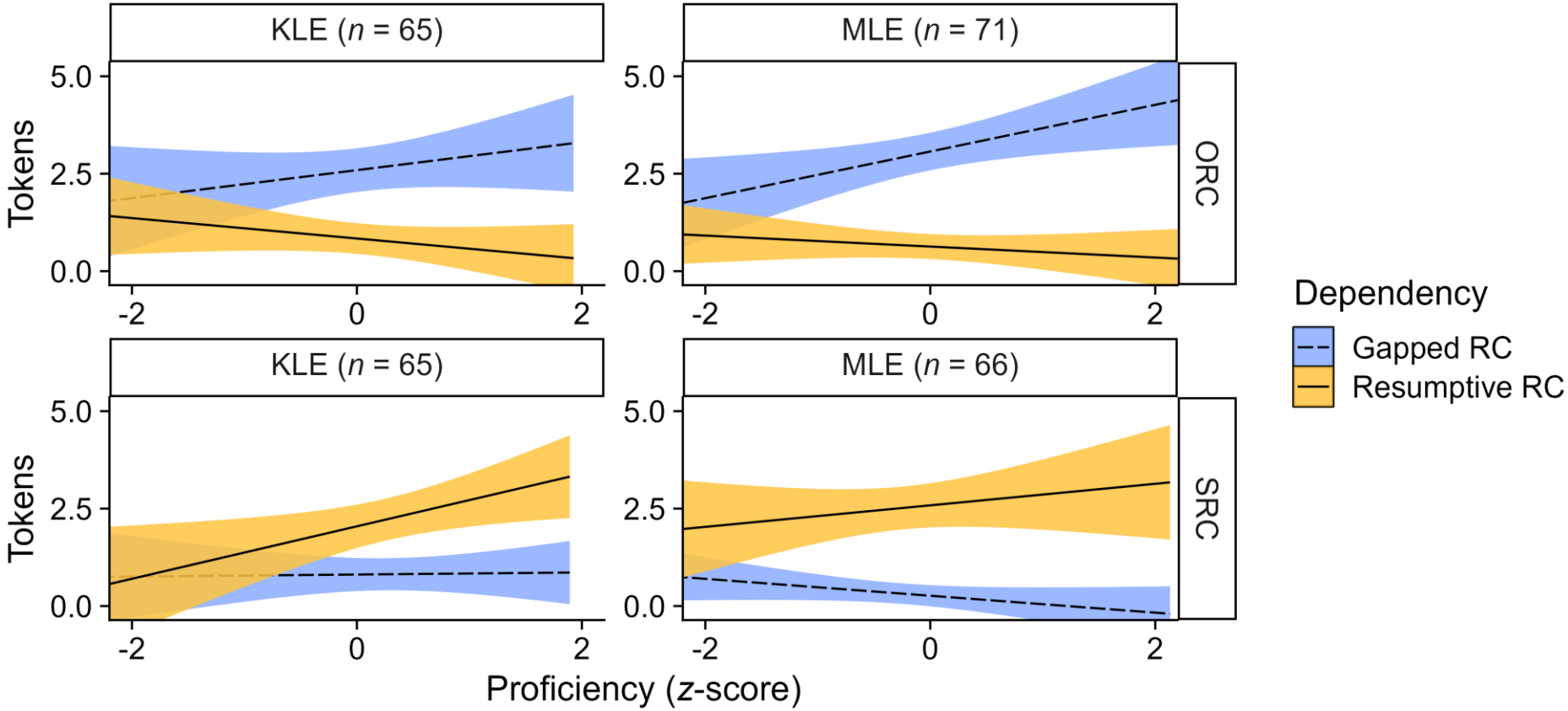
EPT results: Response rates by study and group



Note. Error bars are 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group + (1 | Participant) + (1 | Item).

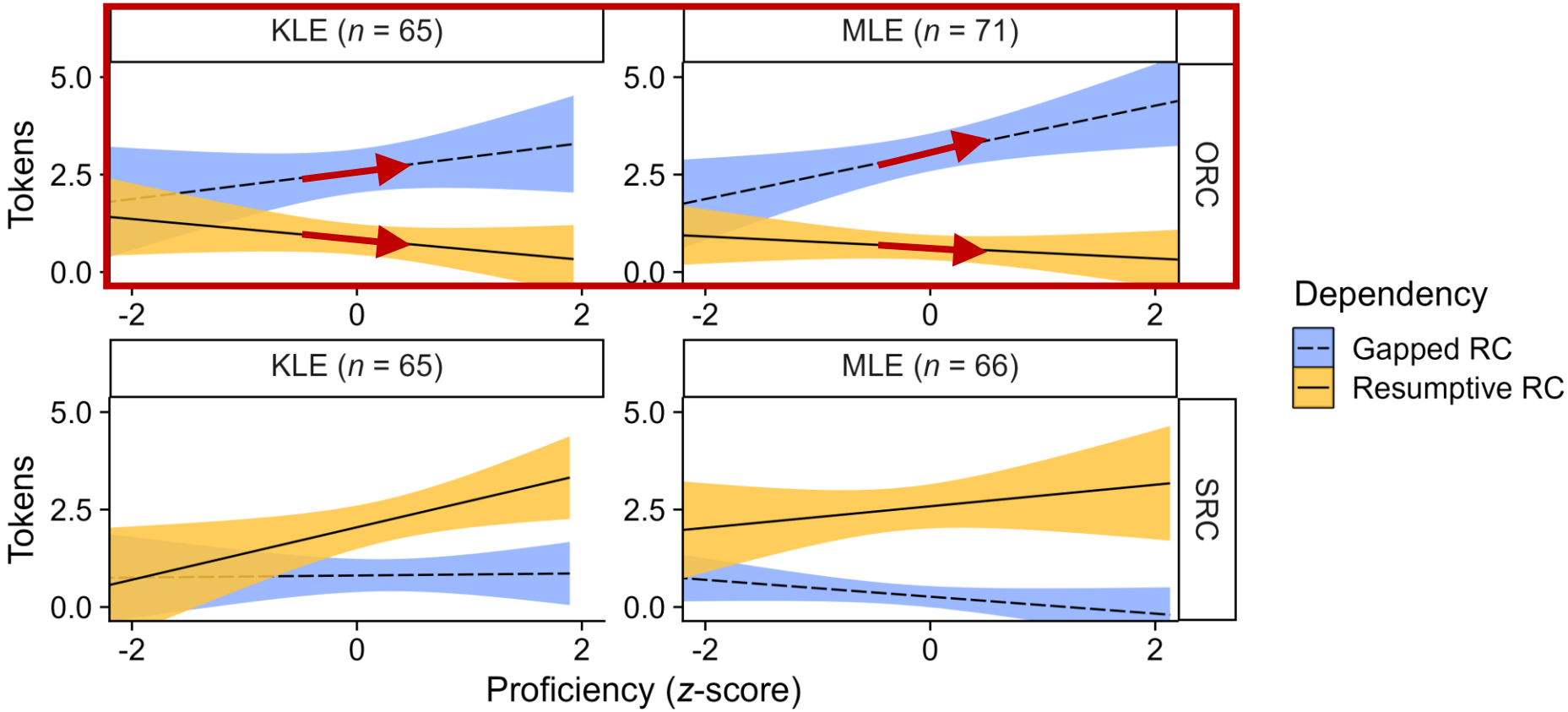
SRC study: Only ENSs produced resumptive RCs significantly more than other response types combined

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group * Proficiency + (1 | Participant) + (1 | Item)

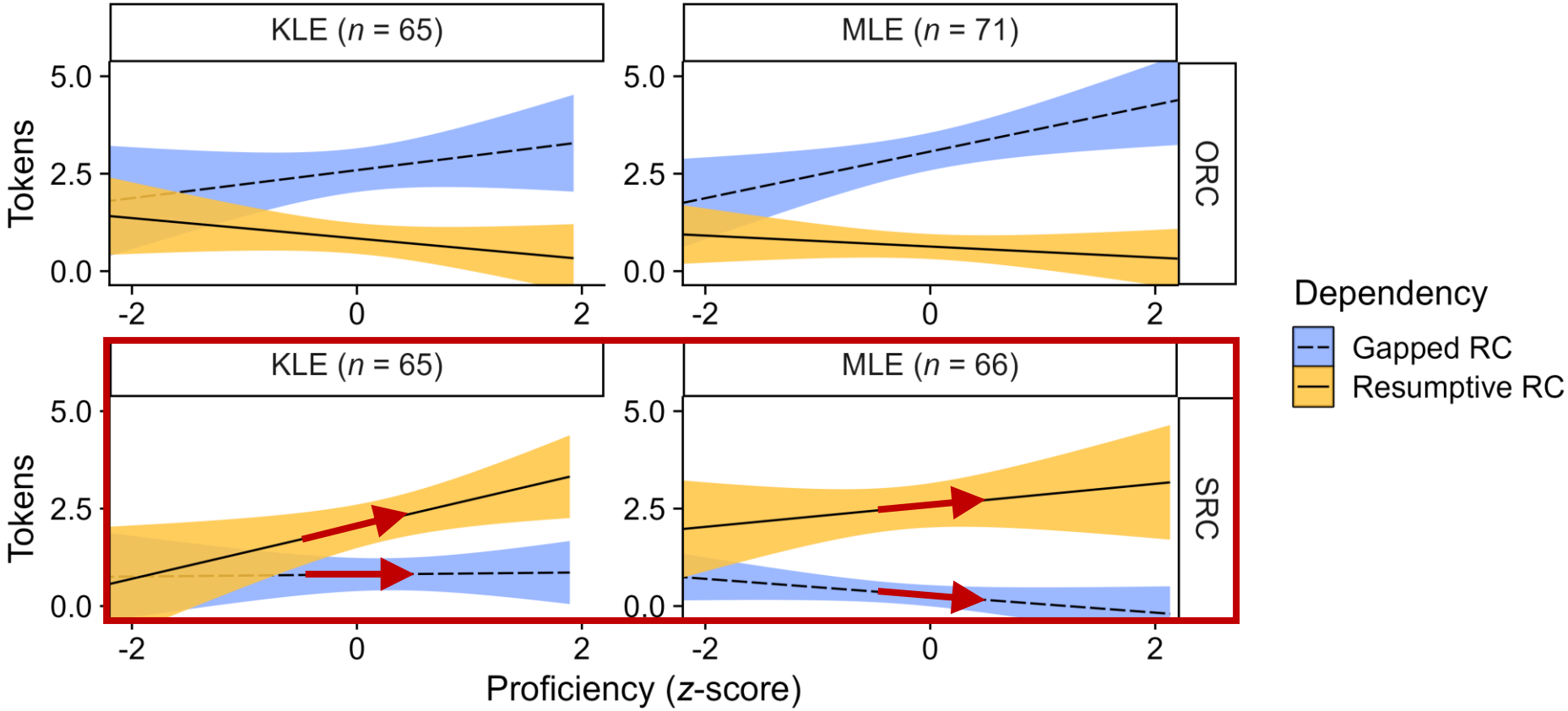
EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group * Proficiency + (1 | Participant) + (1 | Item)

ORC study: With rising proficiency, gapped RCs become more common and resumptive RCs become less common

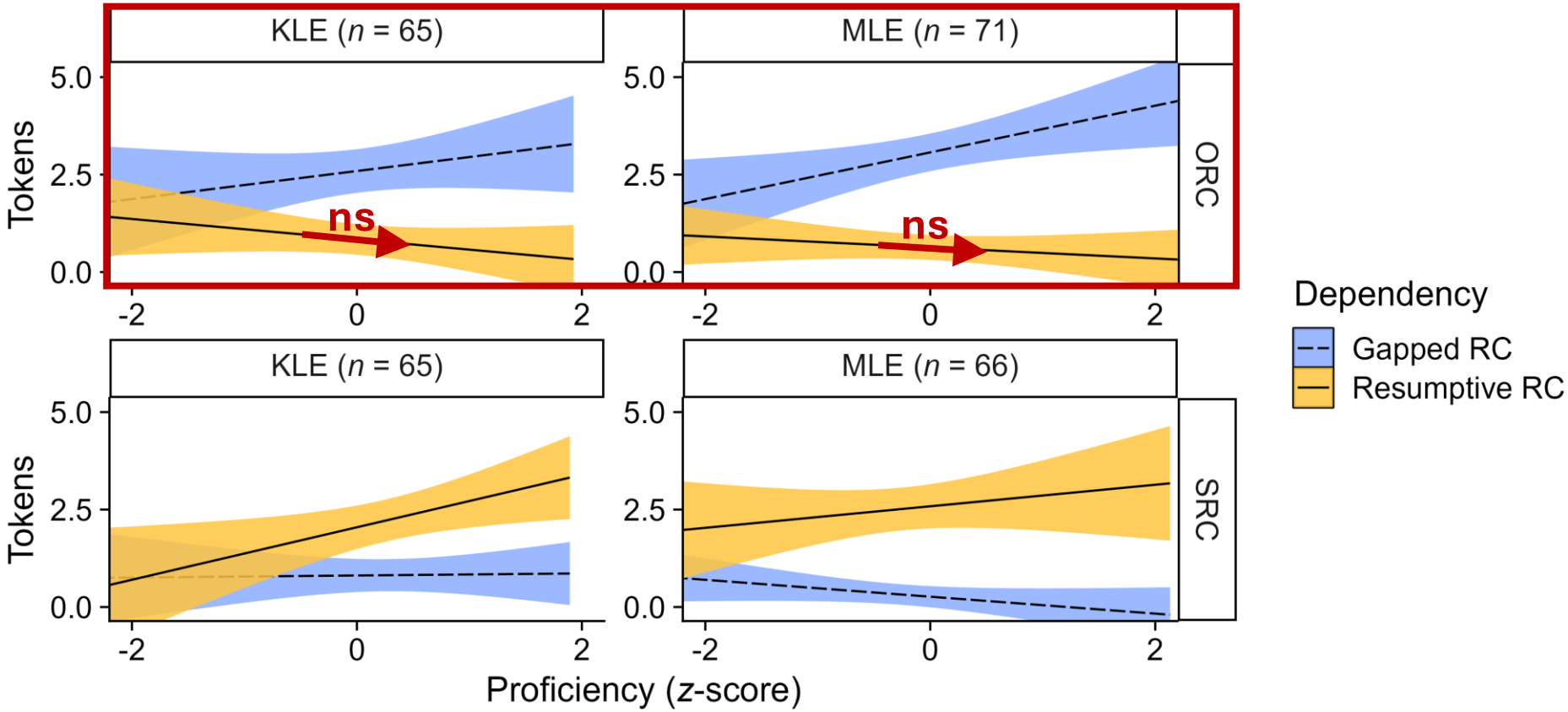
EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group * Proficiency + (1 | Participant) + (1 | Item)

SRC study: With rising proficiency, resumptive RCs become more common and gapped RCs become less common

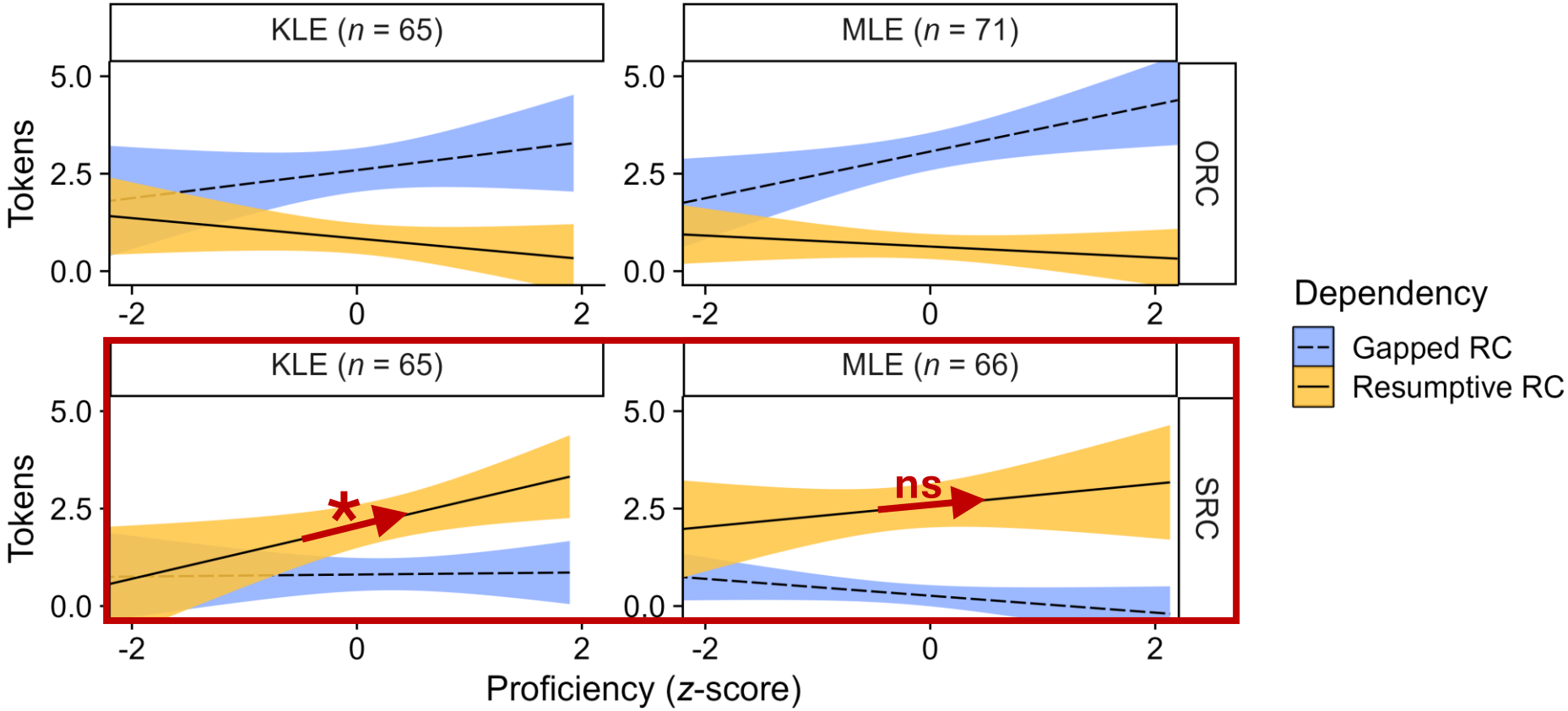
EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group * Proficiency + (1 | Participant) + (1 | Item)

ORC study: No significant relationship between resumption and proficiency

EPT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Logistic mixed-effects model formula: Resumption ~ Group * Proficiency + (1 | Participant) + (1 | Item)

SRC study: Significant relationship between resumption and proficiency for all L2ers and for KLEs considered individually

English acceptability judgment tasks (AJTs)

- 2 × 3 design crossing DEPENDENCY (gapped RC; resumptive RC) and ENVIRONMENT (short-distance; long-distance; *wh*-island)
- 30 critical items (6 conditions × 5 tokens, Latin-squared)
- 42 fillers (28 grammatical; 14 ungrammatical)

Wh-island conditions in the AJTs

Study	Example stimulus with **gapped** vs. **resumptive** RC
ORC	Mary knows the man$_1$ [that I wonder [which officers$_2$ __$_2$ arrested {*__$_1$/***him**$_1$} last week]].
SRC	Mary knows the man$_1$ [that I wonder [which book$_2$ {*__$_1$/***he**$_1$} borrowed __$_2$ last month]].

AJT fillers (*n* = 7 per type)

Type	Example stimulus
F1G	Mary says [this man saw the criminals$_1$ [that ___$_1$ attacked me last week]].
F1U	*This woman says [John met the protesters$_1$ [that ___$_1$ were criticized me last week]].
F2G	This man says [Tina found the article$_1$ [that I published ___$_1$ last year]].
F2U	*Mike says [this woman likes essay$_1$ [that I published ___$_1$ last year]].
F3G	This woman wonders [who$_1$ ___$_1$ saw the journalists$_2$ [that ___$_2$ teased Bill last month]].
F4G	Anna thinks [this man likes the story$_1$ [that she told ___$_1$ last night].

Note. "G" = grammatical; "U" = ungrammatical.

AJT example trial in the SRC *wh*-island condition

stimulus

Mary knows the man that I wonder which book he borrowed last month.

Completely Unacceptable 1 2 3 4 5 6 *Completely Acceptable*

Press the zero key if you cannot rate the sentence

6-point rating scale

I-don't-know option

AJT data exclusions

Experimental error

- ORC study: 1 KLE
- SRC study: 1 MLE

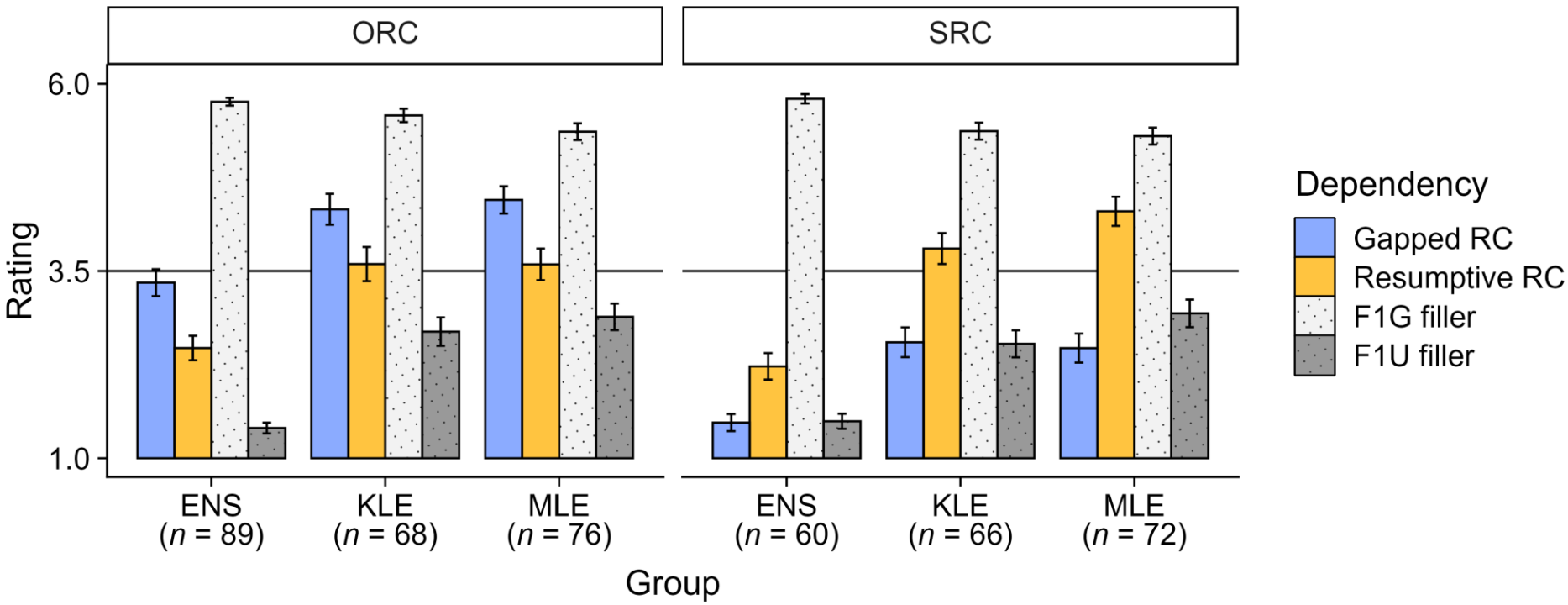
I-don't-know responses

- ORC study: 22 responses (0.13%), 7 from *wh*-island condition
- SRC study: 30 responses (0.21%), 10 from *wh*-island condition

***Yes*-bias outliers**

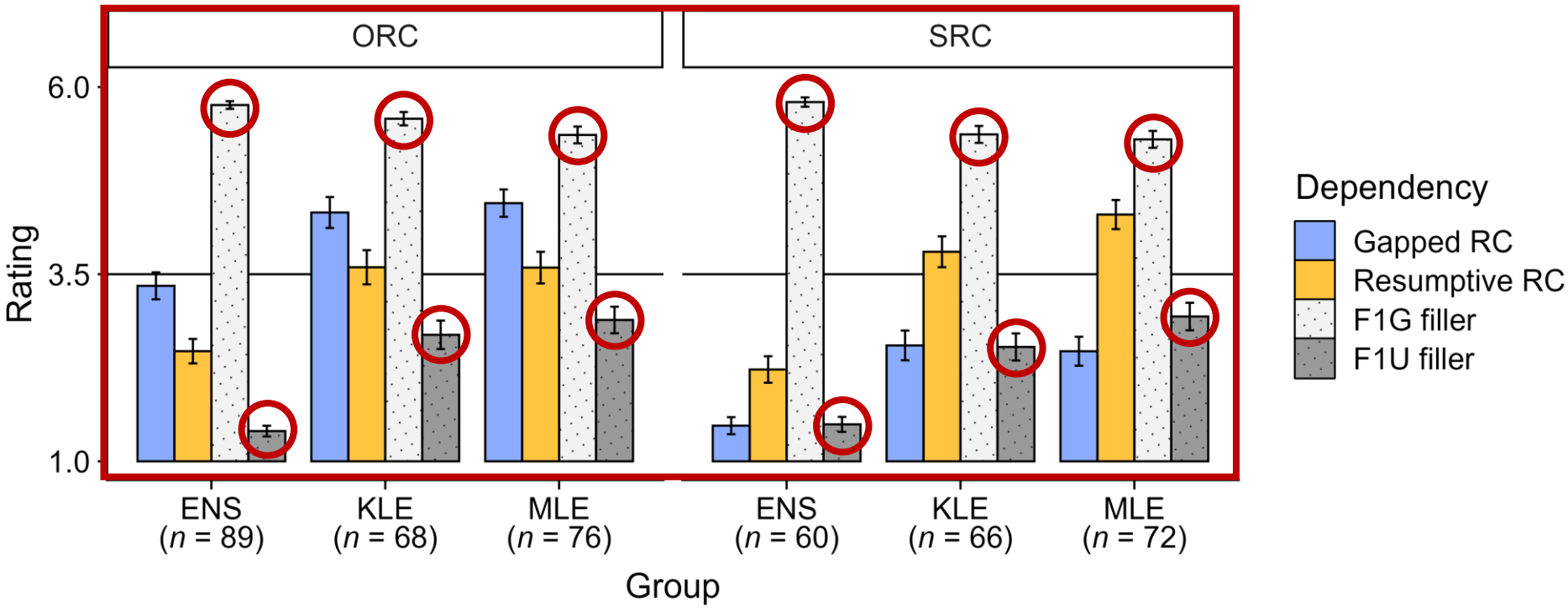
- ORC study: 1 ENS
- SRC study: 1 ENS

AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

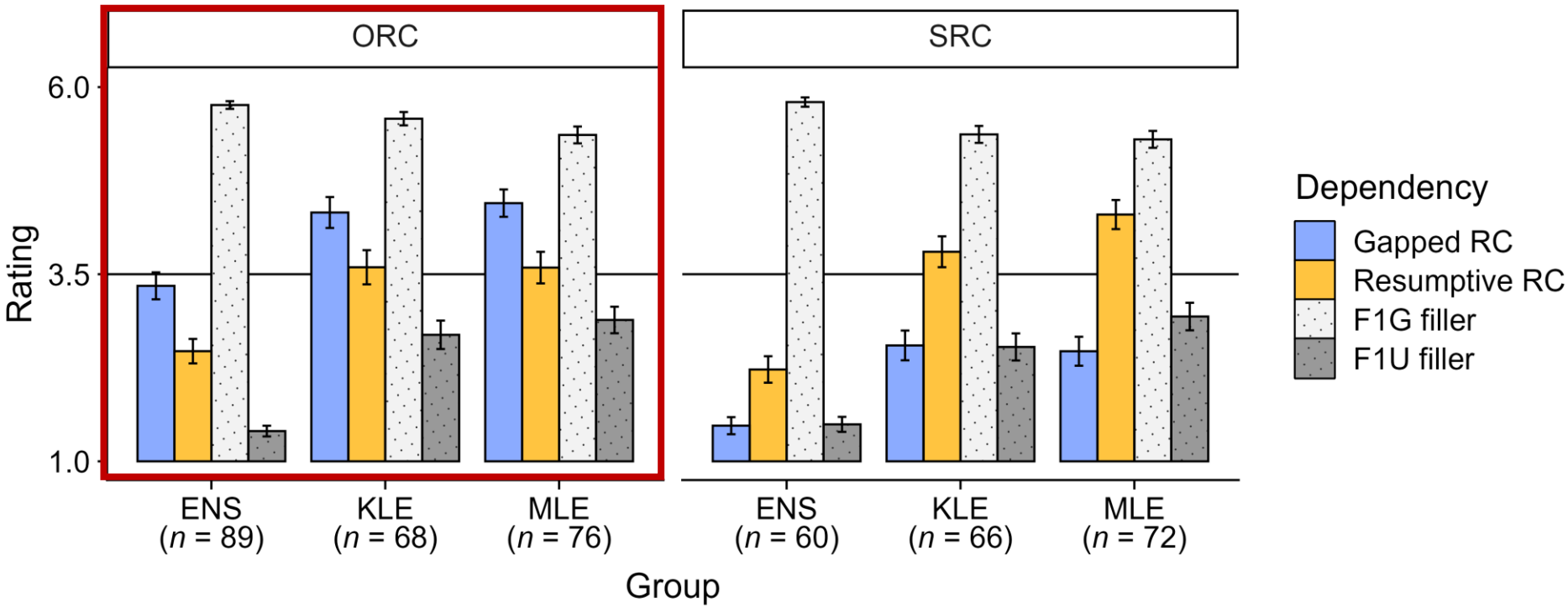
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

Ratings for F1G and F1U fillers patterned very similarly across the two studies

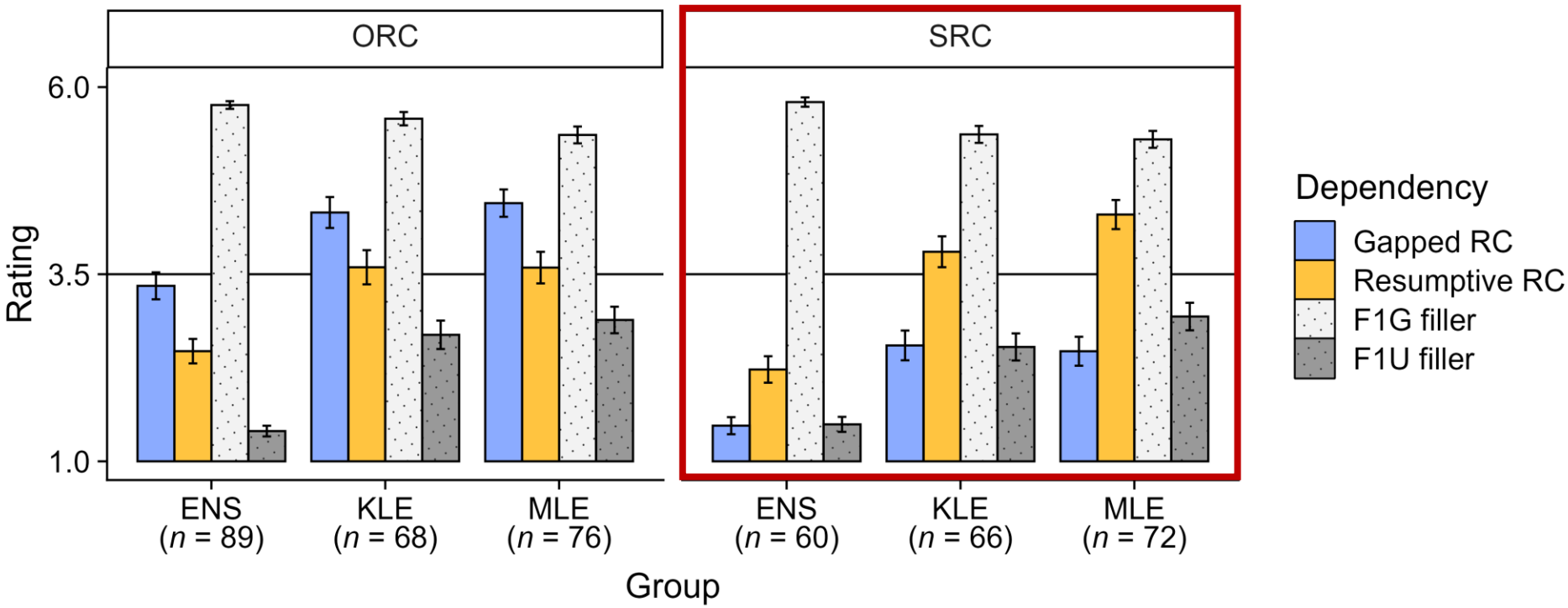
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

ORC study: Gapped RCs higher than resumptive RCs; both intermediate between F1G and F1U fillers

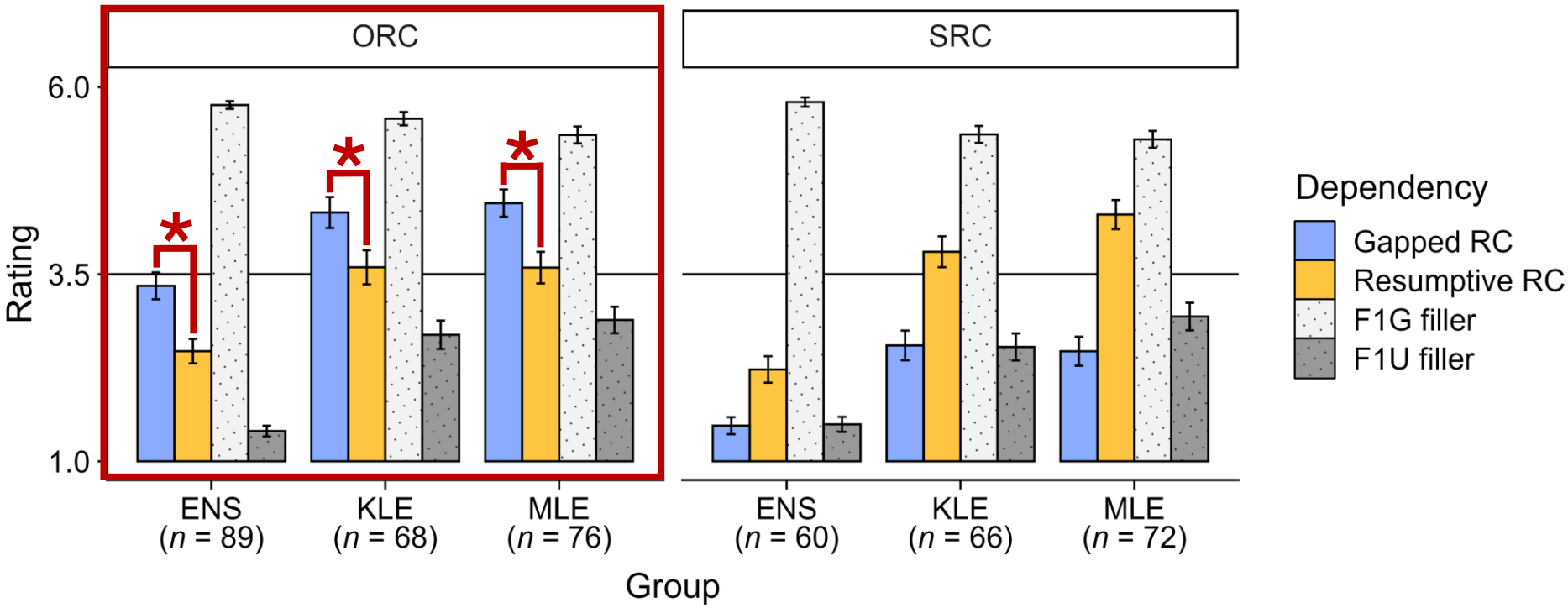
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

SRC study: Gapped RCs lower than resumptive RCs; restricted to lower half of rating scale; on par with or lower than F1U fillers

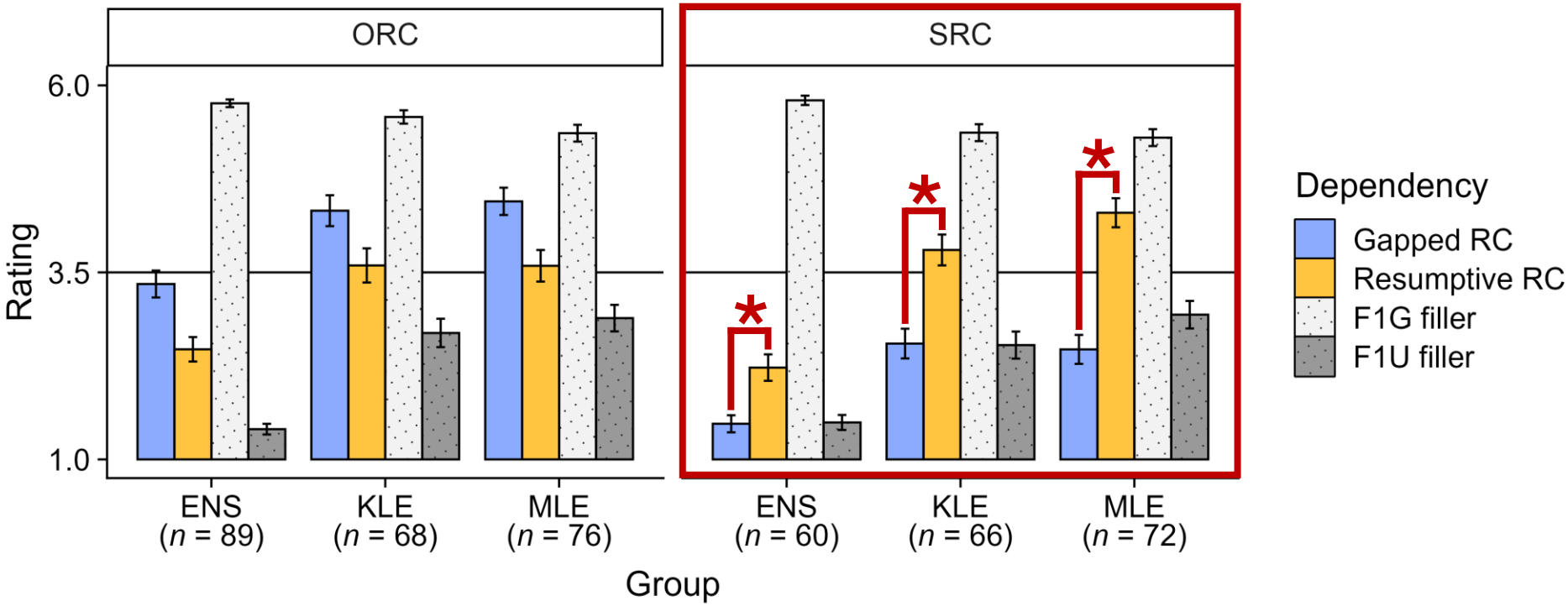
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

ORC study: Gapped RCs significantly higher than resumptive RCs for all groups

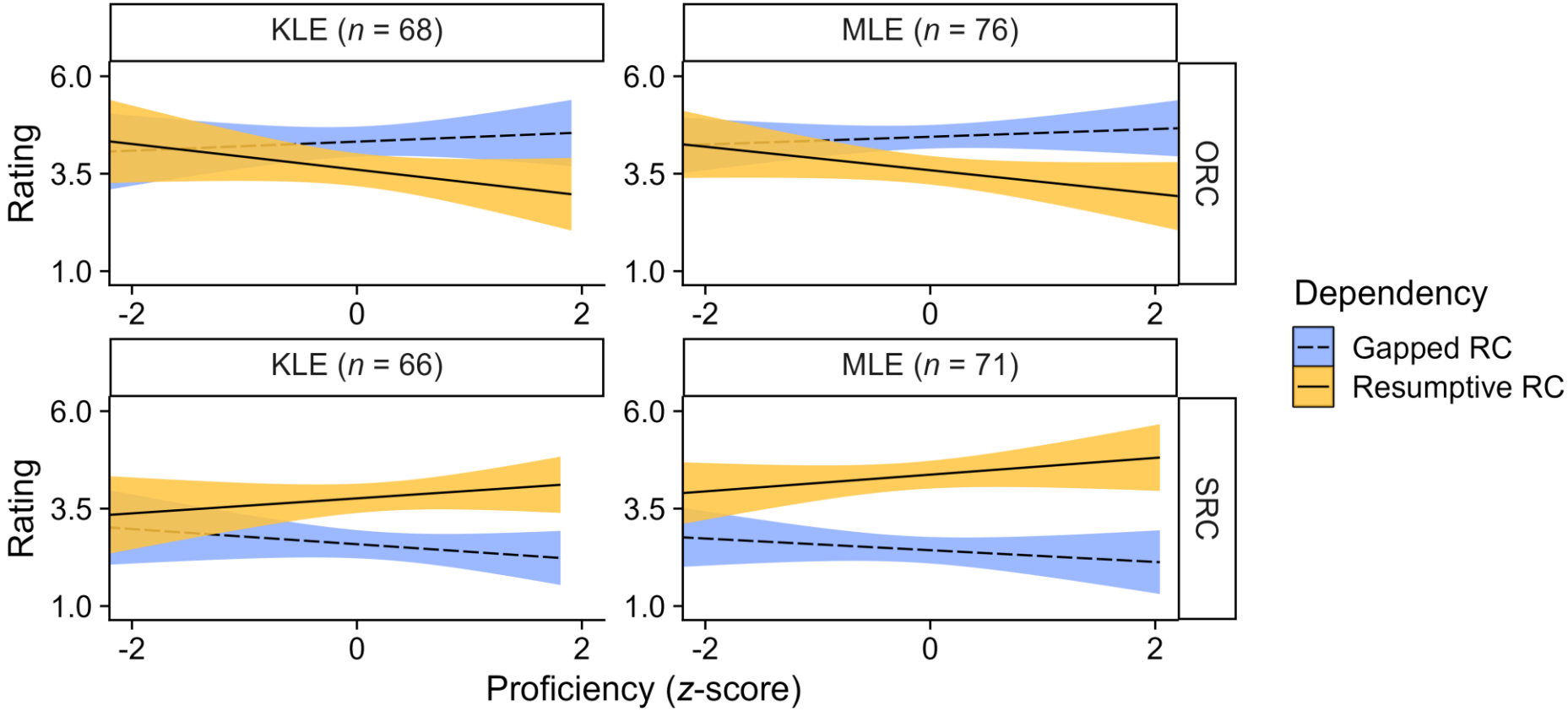
AJT results: Mean ratings by study and group



Note. Error bars are 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group + (1 + Dependency | Participant) + (1 + Dependency + Group | Item).

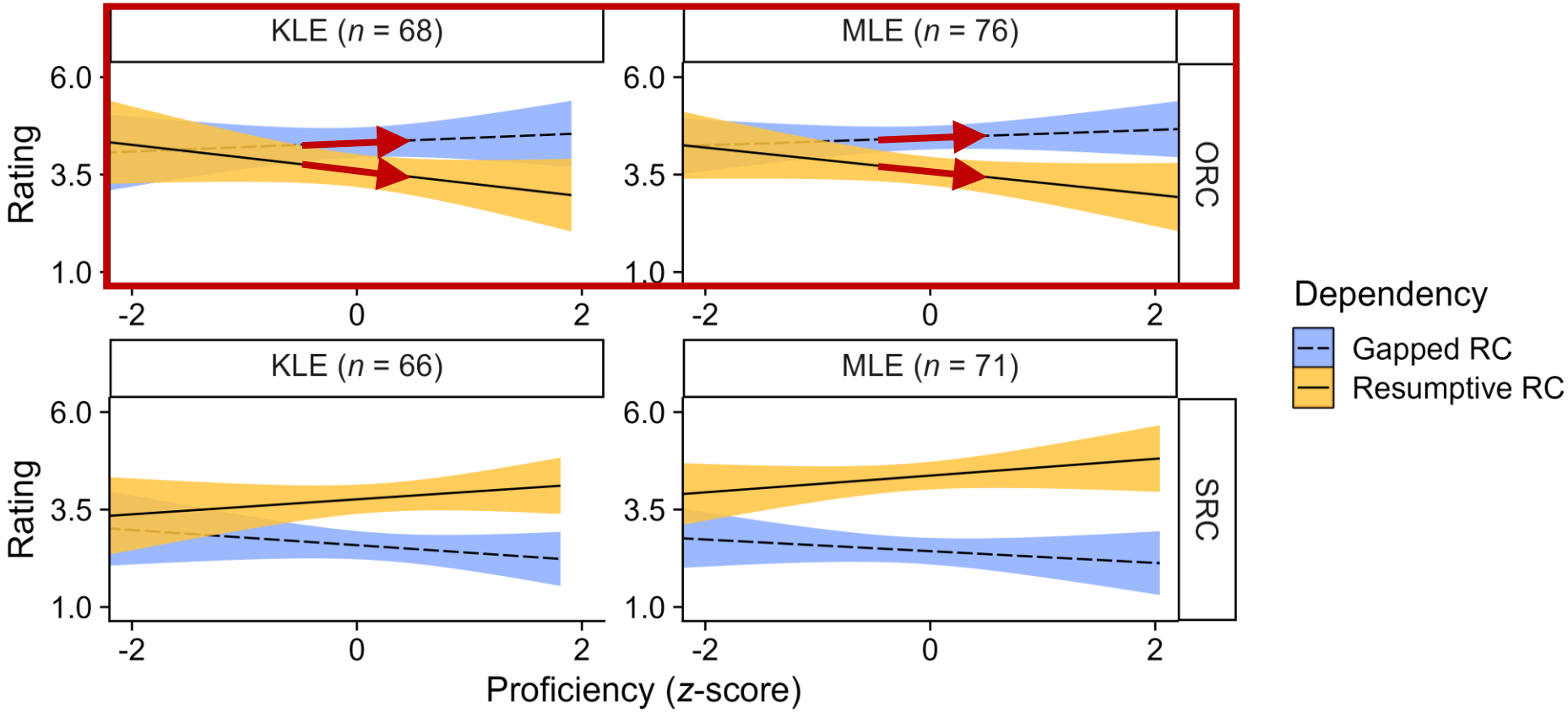
SRC study: Gapped RCs significantly lower than resumptive RCs for all groups

AJT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group * Proficiency + (1 + Dependency | Participant) + (1 + Dependency + Group | Item)

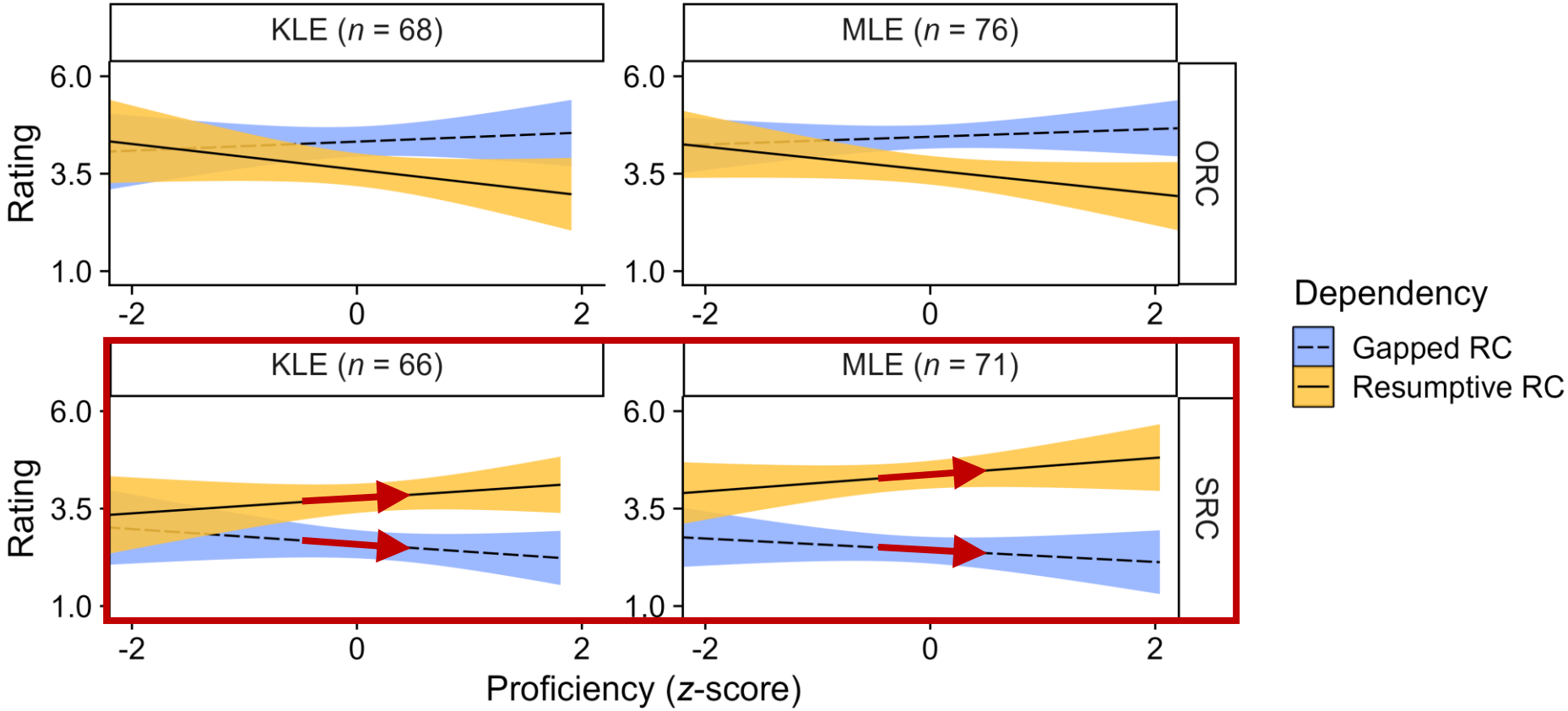
AJT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group * Proficiency + (1 + Dependency | Participant) + (1 + Dependency + Group | Item)

ORC study: Higher proficiency associated with emerging preference for gapped RCs over resumptive RCs

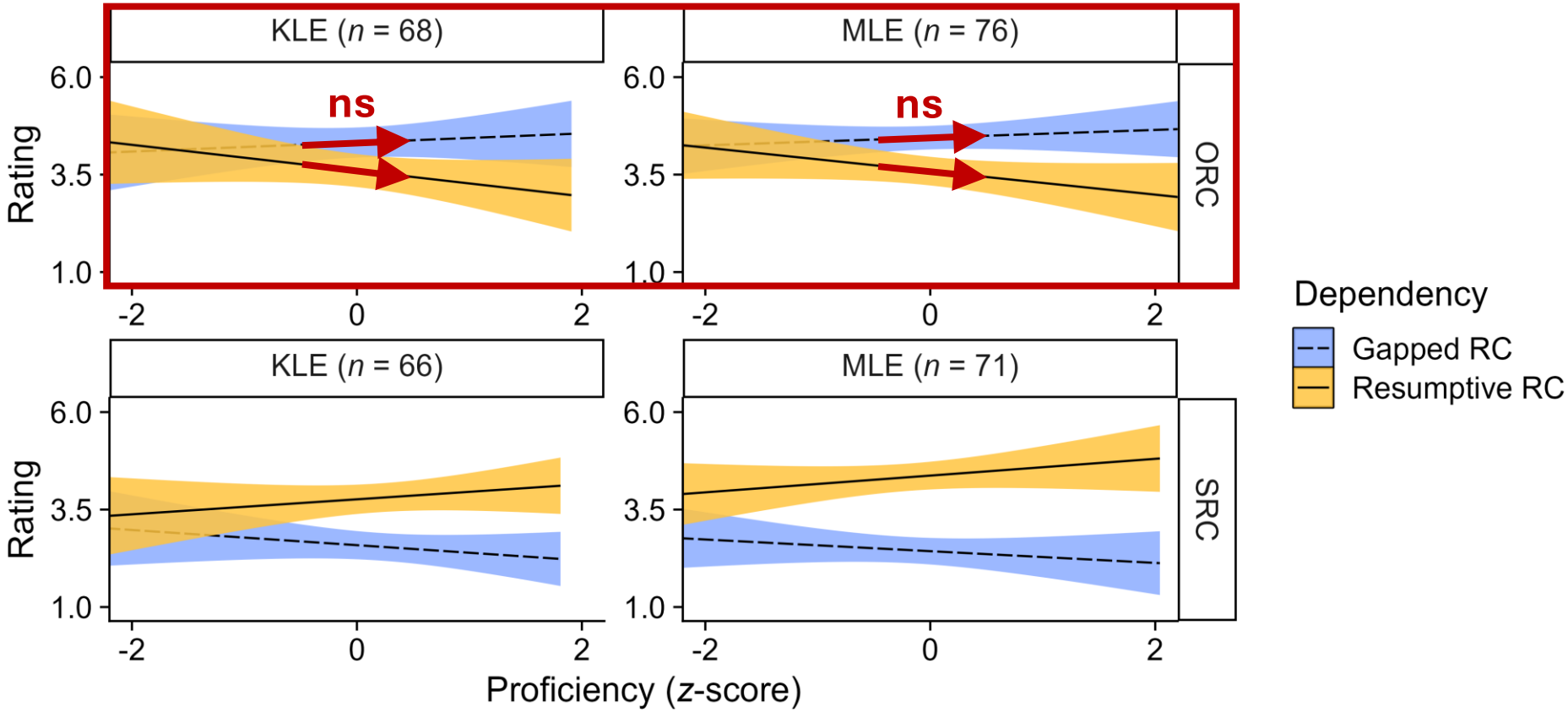
AJT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group * Proficiency + (1 + Dependency | Participant) + (1 + Dependency + Group | Item)

SRC study: Higher proficiency associated with emerging preference for resumptive RCs over gapped RCs

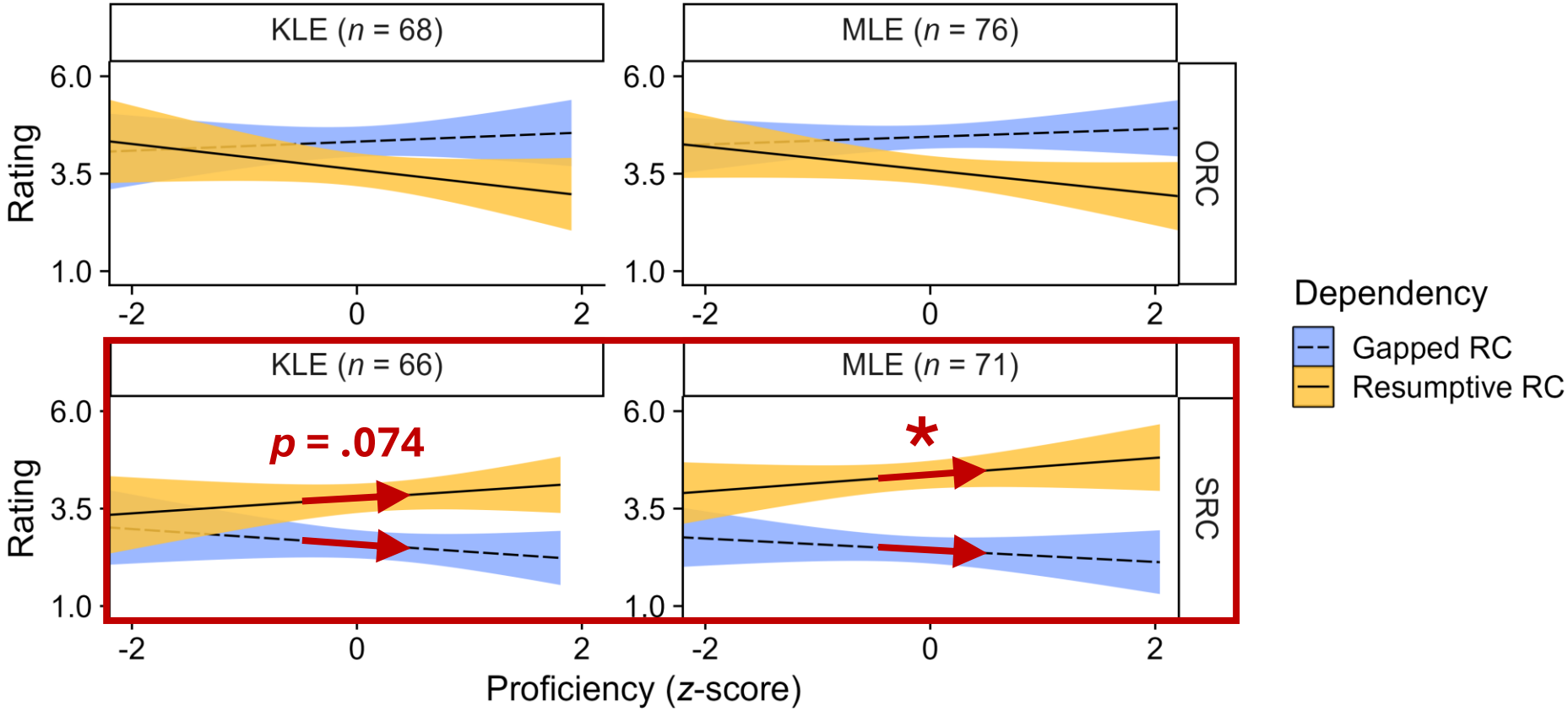
AJT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group * Proficiency + (1 + Dependency | Participant) + (1 + Dependency + Group | Item)

ORC study: Marginal Dependency * Proficiency interaction with all L2ers (p = .085), but not for either group individually (both ps > .10)

AJT results: L2 proficiency effects



Note. Shading marks 95% confidence intervals. Cumulative link mixed model formula: Rating ~ Dependency * Group * Proficiency + (1 + Dependency | Participant) + (1 + Dependency + Group | Item)

SRC study: Dependency * Proficiency interaction significant with all L2ers (p = .005); marginal for KLEs (p = .074), significant for MLEs (p = .025)

Summary of results

***Wh*-island ORCs**

- **All groups:** Gapped RCs produced more often and rated higher than resumptive RCs
- **L2 groups:** Rising proficiency associated with emerging preference for gapped RCs over resumptive RCs

***Wh*-island SRCs**

- **All groups:** Gapped RCs produced less often and rated lower than resumptive RCs
- **L2 groups:** Rising proficiency associated with emerging preference for resumptive RCs over gapped RCs

Research questions revisited

RQ1: Does the subject–object asymmetry in the strength of argument *wh*-islands hold for adult L2 learners of English whose L1 lacks *wh*-movement in *wh*-questions?

Yes, there data from all groups indicated island effects are stronger in *wh*-island SRCs than in *wh*-island ORCs

RQ2: Is the strength of the asymmetry modulated by L2 English proficiency?

Yes, both L2 groups showed an emerging preference for gaps in *wh*-island ORCs and for resumptive nominals in *wh*-island SRCs, bringing them in line with ENSs

Accounting for the asymmetry in island strength

Two major types of explanations:

- Representational accounts
- Processing accounts

Representational account: ECP violation

Subject-gap islands violate an extra grammar constraint, such as the Empty Category Principle (ECP)

ECP: Traces must be properly governed (Chomsky, 1981)

(5) *the intern$_1$ [OP$_1$ that I wonder [who$_2$ t_2 hired t_1]]

✓ verb *hired* c-commands the object trace t_1

✓ no barriers intervene between *hired* and t_1

Proper government achieved in *wh*-island ORCs

(6) *the customer$_1$ [OP$_1$ that I wonder [what$_2$ t_1 bought t_2]]

✓ antecedent OP$_1$ c-commands subject trace t_1

✗ *wh*-phrase *what*$_2$ is a barrier between OP$_1$ and t_1

Proper government fails in *wh*-island SRCs

(McDaniel & Cowart, 1999; Keffala & Goodall, 2011)

Processing account: Clause planning

- Clause planning and filler integration → spikes in cognitive load
- Processing load more distributed across clause in object-gap islands than in subject-gap islands
- Overlapping spikes → acute processing difficulty

(7) RC with object gap in *wh*-island (easier to process)

*the intern [that I wonder [who hired last week]]

XXXXXXX ← clause-planning spike

XXXXXXX ← filler-integration spike

(8) RC with subject gap in *wh*-island (harder to process)

*the customer [that I wonder [what bought last week]]

XXXXXXX ← clause-planning spike

XXXXXXX ← filler-integration spike

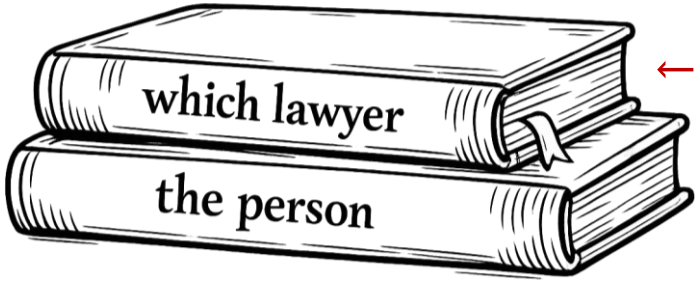
(McDaniel et al., 2015)

Processing account: Push-down storage

- **Push-down storage principle:** Recently stored items accessed more readily than previously stored ones (Marcus, 1978)
- *Wh*-island ORCs → top item retrieved first (easier)
- *Wh*-island SRCs → bottom item retrieved first (harder)

(9) a. *the person$_1$ [that I wonder [which lawyer$_2$ __$_2$ hired __$_1$]] (*wh*-island ORC) >

b. * the person$_1$ [that I wonder [which lawyer$_2$ __$_1$ hired __$_2$]] (*wh*-island SRC)



← top of memory stack

← bottom of memory stack

(O'Grady, 2005)

Limitations and future directions

1. Between-subjects design prevented direct statistical comparisons between ORCs and SRCs
   - ▶ Use within-subjects designs where possible
2. Results equally compatible with representational accounts and processing accounts
   - ▶ Test whether inserting sentential adverbial after *wh*-phrase makes *wh*-island SRCs better (see Culicover, 1993)

(10) a. *the customer$_1$ [that I wonder [when$_2$ **exactly** __$_1$ bought the tickets __$_2$]] >
b. * the customer$_1$ [that I wonder [when$_2$ __$_1$ bought the tickets **exactly** __$_2$]]

Conclusion

- Island effects for argument *wh*-islands are stronger in SRCs than in ORCs for ENSs, KLEs, and MLEs
- L1-Korean and L1-Mandarin L2 learners of English can become sensitive to the asymmetry even without help from the L1
- The difference in island strength comes into sharper focus with rising L2 proficiency, bringing KLEs and MLEs in line with ENSs
- More research needed to determine whether representational accounts or processing accounts provide the best explanation

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Software

Experimental software:

The experiments were built using the jsPsych library (de Leeuw, 2015) and hosted on Cognition (https://www.cognition.run). Participants were recruited on Prolific (https://www.prolific.com), on Korean online job boards such as SNULife (https://www.snulife.com) and Spec Up (https://cafe.naver.com/specup), on Taiwan's PTT bulletin board system (https://www.ptt.cc), and through the Language Analysis and Experimentation Labs at the University of Hawaiʻi at Mānoa (https://sites.google.com/a/hawaii.edu/laelabsedit).

Data analysis tools:

All analyses were performed in R (R Core Team, 2024). Mixed-effect models were fit using the *lme4* (Bates et al., 2017) and *ordinal* (Christensen, 2019) packages. Post-hoc testing was done with the *emmeans* package (Lenth, 2022).

Image credits:

This presentation uses images generated in ChatGPT (https://chatgpt.com) and icons from Flaticon (https://www.flaticon.com).

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감사합니다 谢谢

Thank you!

Mahalo Danke

fredzenker.github.io/gasla2026.pdf

Appendix

EPT all conditions

Study	Environment	Target response with **gapped** vs. **resumptive** RC
ORC	Short-distance	the man$_1$ [that the officers arrested {__$_1$/***him**$_1$} last week]
ORC	Long-distance	the man$_1$ [that Mary thinks [the officers arrested {__$_1$/***him**$_1$} last week]]
ORC	*Wh*-island	the man$_1$ [that Mary wonders [which officers$_2$ __$_2$ arrested {*__$_1$/***him**$_1$} last week]]
SRC	Short-distance	the man$_1$ [that {__$_1$/***he**$_1$} borrowed a book last month]
SRC	Long-distance	the man$_1$ [that Mary thinks [{__$_1$/***he**$_1$} borrowed a book last month]]
SRC	*Wh*-island	the man$_1$ [that Mary wonders [which book$_2$ {__$_1$/***he**$_1$} borrowed __$_2$ last month]]

AJT all conditions

Study	Environment	Target response with **gapped** vs. **resumptive** RC
ORC	Short-distance	Mary knows the man$_1$ [that these officers arrested {__$_1$/***him**$_1$} last week]
ORC	Long-distance	Mary knows the man$_1$ [that I think [these officers arrested {__$_1$/***him**$_1$} last week]]
ORC	*Wh*-island	Mary knows the man$_1$ [that I wonder [which officers$_2$ __$_2$ arrested {*__$_1$/***him**$_1$} last week]]
SRC	Short-distance	Mary knows the man$_1$ [that {__$_1$/***he**$_1$} borrowed this book last month]
SRC	Long-distance	Mary knows the man$_1$ [that I think [{__$_1$/***he**$_1$} borrowed this book last month]]
SRC	*Wh*-island	Mary knows the man$_1$ [that I wonder [which book$_2$ {__$_1$/***he**$_1$} borrowed __$_2$ last month]]

AJT procedure for identifying *yes*- and *no*-bias outliers

***Yes* bias**

- Mean rating above 3.5 and in excess of 3 SDs above group mean on ungrammatical trials in both critical items and fillers

***No* bias**

- Mean rating below 3.5 and in excess of 3 SDs below group mean on grammatical trials in both critical items and fillers